

设备选型

基本要求：

Kill Switch (Emergency Stop) Requirements

On-Board: A hard-wired, on-board, emergency stop system.

Off-Board: A wireless remote emergency stop, located off-board and on its own frequency and link.

Each uncrewed system must carry an **emergency recovery light** that operates independently from all other platform power systems and is capable of functioning underwater for at least 12 hours.

USV

Requirements:

Kill Switch (Emergency Stop) Requirements

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Off-Board: A wireless remote emergency stop, located off-board and on its own frequency and link.

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Size: The USV must fit within a **2m x 1m x 1m** (L x W x H) "box".

Weight: The entire maritime system must weigh less than or equal to **60kg**.

Buoyant: The USV must be **positively buoyant** and hydrodynamically stable enough to operate safely in autonomous and manual modes, as well as when being towed during competition.

Visual Feedback System: The USV must have an onboard **visual feedback system** to clearly indicate the operational status of the system to improve the safety of competition operations.

All propellers must be shrouded.

1. Blue boat 120cm*93cm*38cm

2 battery 18hrs

Raspeberry pi based flight controller runs arduRover

2.4GHz Telemetry

M9N GPS

<https://bluerobotics.com/store/boat/blueboat/blueboat/>
bluerobotics.com

2. 南京超维

https://item.taobao.com/item.htm?id=802746017233&pisk=ge4g1bwa6v9WjlpFrKu1vS21mZBp1Vgj_-LxHqHVVykBW-3tkX...
item.taobao.com

<https://cwkj-tech.yuque.com/bsge84/suv1/xgqa5l4yo0zwnmt7>
免责声明 · 超维空间科技
注意!!! 为保护用户的合法权益，请您在使用本产品前务必...

配件清单：

产品名称	数量	单价	总价	淘宝价格	
1.2米双体船船体	1	7600	7600		
电机	2	3500	7000		
电调	2	1776	3552		
24V 20AH 电池（具体型号待确认）	2	1200	2400		
24V充电器	2	800	1600		
云卓G20 遥控器	1	3988	3988	3199	

					 Skydroid- G20 User manual V1.0  Airmobi Limited - www.airmobi.com  G20遥控器与GR01接收机.pdf
pixhawk6 c 飞控	1	4000	4000		
UM982 RTK	1	2000	2000		
BT-560天 线	1	360	360		
基站端 (F9P+BT -770天 线)	1	2140	2140		
Jetson orin nx 16G模块 (自己提 供)	1	6500	6500	自己提 供	
D435相机	1	2499	2499		
Mid360S 三维激光 雷达	1	5000	5000		

915Mhz的 数传 (自己 买)	1	1000	1000	自己提 供	
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USV 额外配件

喷水云台

宏烨 马宏烨 这里写云台+喷水

瞄准云台：

竞赛版 【YT260-U28H】 12v/无刷总线舵机 + 赠12V 3A电源



demo演示：



https://item.taobao.com/item.htm?abbucket=9&id=619989912076&mi_id=0000jjXx-BQhzHwZO9ED_fP1oYZQx7wcivSJFS...
[item.taobao.com](https://item.taobao.com/item.htm?abbucket=9&id=619989912076&mi_id=0000jjXx-BQhzHwZO9ED_fP1oYZQx7wcivSJFS...)

摄像头：OAK D LITE 定焦款（开票）

https://item.taobao.com/item.htm?abbucket=9&id=876346135136&mi_id=0000bSXGfTH8ag7Fq2qckpB3e7BWlilmydWD...
[item.taobao.com](https://item.taobao.com/item.htm?abbucket=9&id=876346135136&mi_id=0000bSXGfTH8ag7Fq2qckpB3e7BWlilmydWD...)

喷水装置：

带调速自吸泵+过滤+3米管

https://item.taobao.com/item.htm?abbucket=9&id=870204615636&mi_id=00008I9siyfrsyjKM80E9J8UXiSbIIU0ju_haaxe...
[item.taobao.com](https://item.taobao.com/item.htm?abbucket=9&id=870204615636&mi_id=00008I9siyfrsyjKM80E9J8UXiSbIIU0ju_haaxe...)

演示视频：



喷嘴6mm可调-0.8mm孔

https://item.taobao.com/item.htm?abbucket=9&id=898801724455&mi_id=0000sxgLY_mtpCatKuDxtygJi-16RJjNRxa_4cwViKJdqY4&ns=1&priceTId=2150460017858483625164889e11c9&skuId=5918640086463&spm=a21n57.1.hoverItem.6&utparam=%7B%22aplus_abtest%22%3A%2256934fb9117c98a12d3e74d010f18c8a%22%7D&xxc=taobaoSearch

PG8-6(8转6) 5只装 变径直通接头

https://item.taobao.com/item.htm?abbucket=9&id=744450185520&mi_id=0000mXAwY2WTwyyFFNWA4V5E7fTOC-B1Xicq...
item.taobao.com

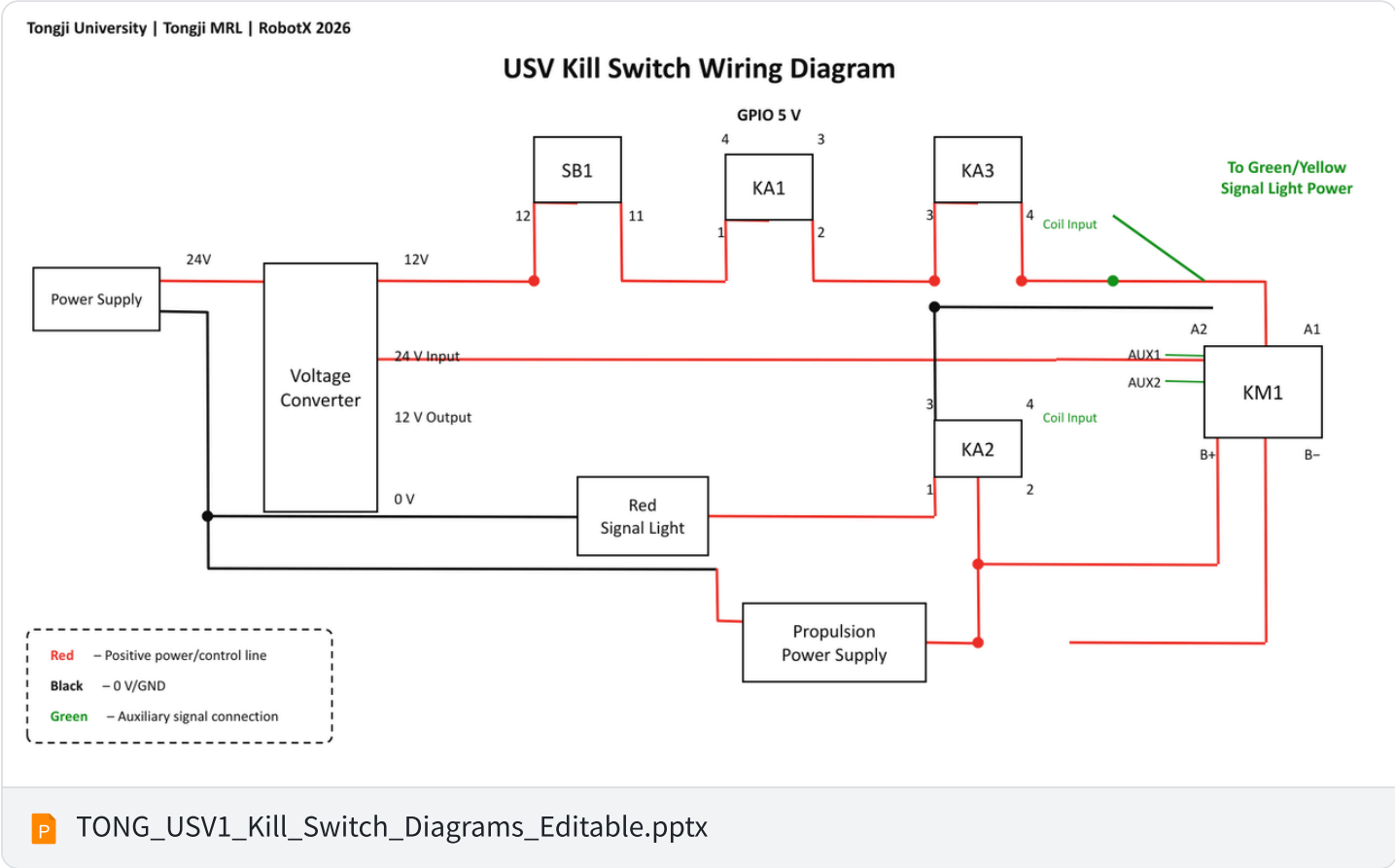
304材质 $\Phi 8\text{mm}$ 含垫片*2

https://item.taobao.com/item.htm?abbucket=9&id=835053473188&mi_id=0000M1UCJ8GRJ93eEXI2sdCkULgxoFpGLLjnYa...
item.taobao.com

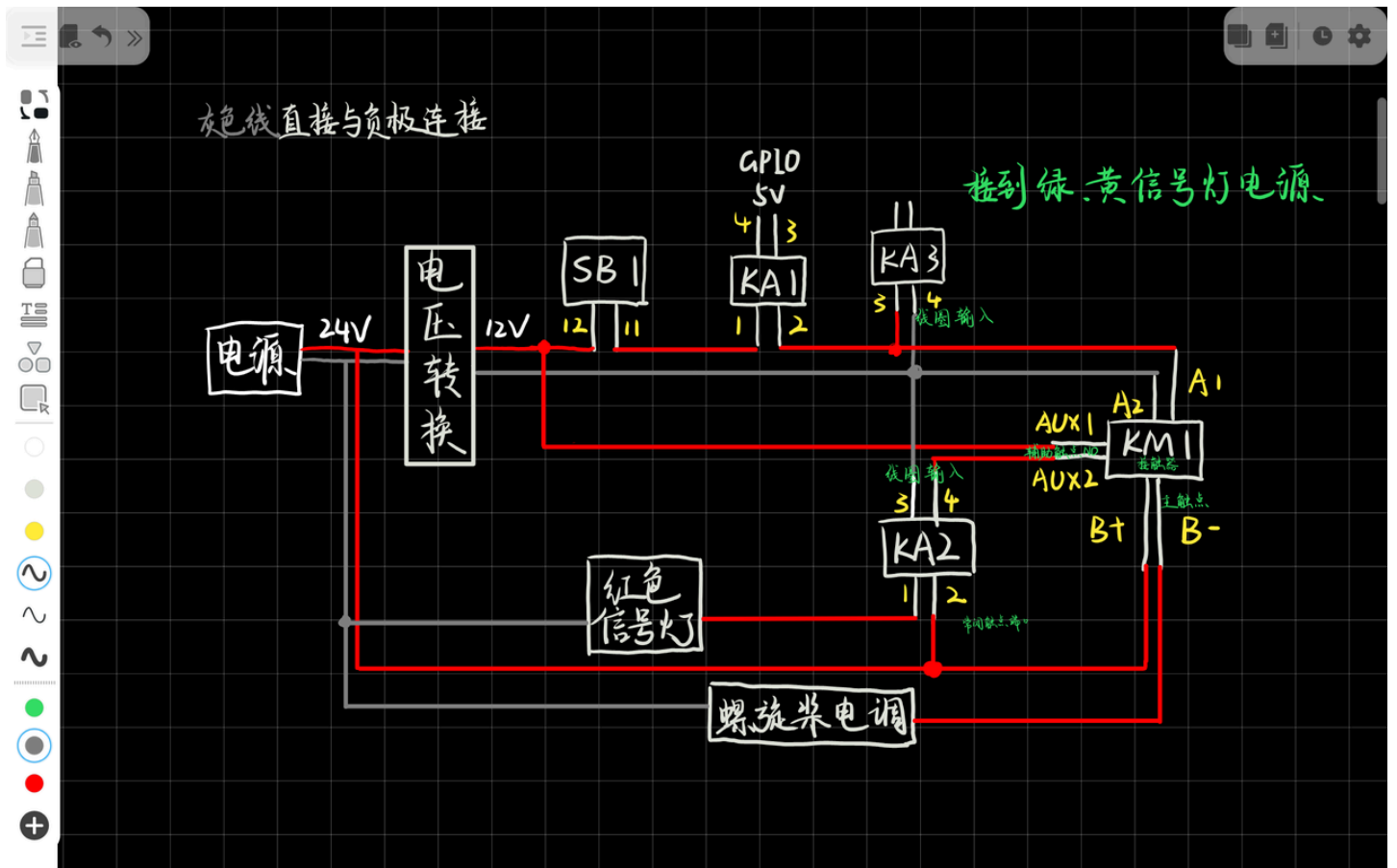
状态指示灯

https://shop318867446.taobao.com/category.htm?spm=pc_detail.30350276.shop_block.dshopinfo.6ad57dd65L83mv
shop318867446.taobao.com

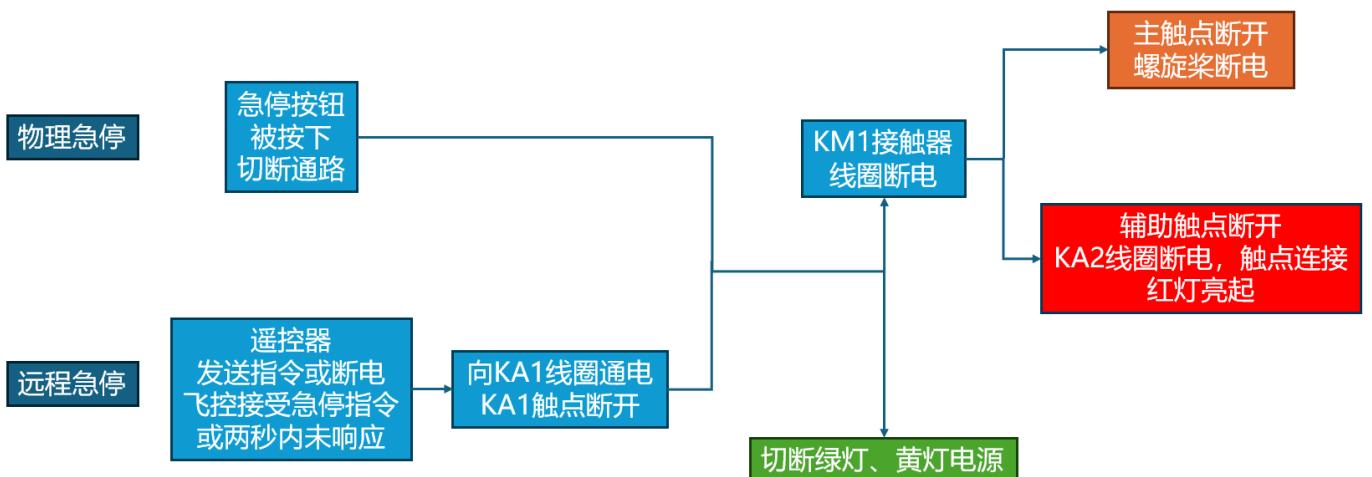
E-stop system



电路连接图



物理急停与远程急停触发原理



元器件选型

名称	型号	价格	链接	

电压转换	24v转12v直流转换器 5A小铝	25	item.taobao.com
KA1、KA2	SUMGLE常闭型固态继电器5-24VDC 10A DL-D0710M-B	63 (31.5*2)	item.taobao.com
KA3	常开型固态继电器SSR-25DD	8	item.taobao.com
KM1	直流接触器EV200HAANA	140	item.taobao.com
SB1	急停按钮XB2BS542C	24.5	item.taobao.com

UAV

不同电量下续航时间估算：

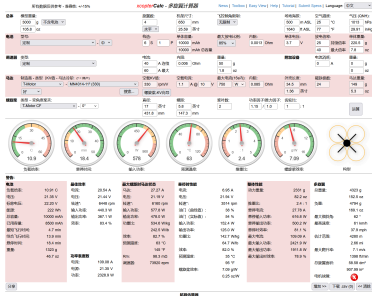
格式电池Lipo

12000mAh 6S 22.2V

格氏正品
现货发售

尺寸(MM)：195(长)X70 (宽) X54 (厚)

重量：1523g

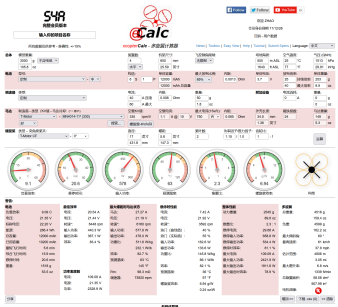


12000mAh 6S 22.2V

格氏正品
现货发售

尺寸(MM)：195(长)X70 (宽) X54 (厚)

重量：1523g



14000mAh 6S 22.2V

格氏正品
现货发售

尺寸(MM)：235 (长)X70 (宽) X53 (厚)

重量：1868g



16000mAh 6S 22.2V

格氏正品
现货发售

尺寸(MM) : 190(长)X86 (宽) X60 (厚)

重量 : 1925g



(标准款) 22000mAh 6S 22.2V

尺寸(MM) : 215(长)X100 (宽) X72 (厚)

重量 : 2816g

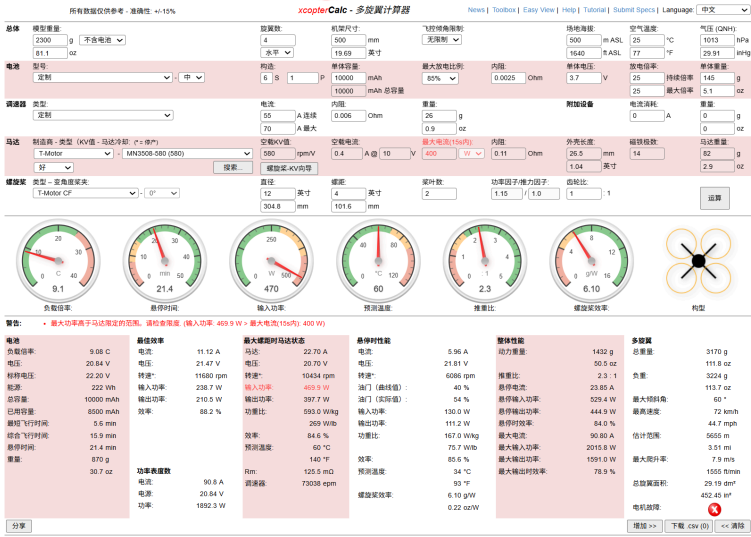


iFlight Li-ion

产品参数

电芯	EVEINR21700-40PL	EVEINR21700-50PL	EVEINR21700-40PL	EVEINR21700-50PL
容量	4000mAh	5000mAh	8000mAh	10000mAh
电压	22.2V	22.2V	22.2V	22.2V
电芯组合	6S1P	6S1P	6S2P	6S2P
瓦时(Wh)	88.8Wh	111Wh	177.6Wh	222Wh
尺寸(mm)	43*63*74	43*63*75	43*64*147	43*63*148
重量(g)	430±5g	436±5g	840±10g	855±10g
放电倍率	17.5C	25C	17.5C	25C
放电插头	XT60H-F	XT60H-F	XT60H-F	XT60H-F

注：页面所展示的尺寸和重量，仅供参考。实际数据可能会因为产品更新、不同批次或个体差异等原因造成变化，具体请以实际为准。



机架

https://item.taobao.com/item.htm?id=641264949315&mi_id=00007JwT4-xz1ZMPBqrg9gpuPMjR2XSxjxAzsR7I5wYOD0s&s...

item.taobao.com



机架	Holybro x500	1600	
电机支架	Tarot 16mm支架*6	300	
飞控	Holybro Pixhawk 6x mini	2000	
分电板	Holybro PM03D	300	
电动脚架	自动收放起落架	200	
电机	T-motor MN3508 *6	2400 (399*6)	
ESC	T-motor Air 40A *6	1380 (229*6)	
桨	T-motor P12*4碳纤*4对	1000 (249*4对)	
电池	iFlight Fullsend LR 6s2p 10000mah*2	1200 (600*2)	
锂电池防爆袋	iFlight翼飞 锂电池防爆袋	120 (60*2)	
RTK(暂时不用)		2200	

	Holybro H-RTK ZED-F9P DroneCAN		
GPS	M9N GPS		
jetson载板	C2401控元	2000	
Jetson orin	Jetson orin nx/nano	--	
激光测距（近距离夹取时测距）	Benewake TFmini-S	270	
前下视相机（远距离搜索）	Arducam B0495 2.3MP AR0234 Color Global Shutter USB 3.0	1500	
下视相机	OAK-D Lite	1500	
数传（RTK）	P900*3	4500（1500*3）	
遥控接收	Radiomaster RP3接收机	120	
数传（QGC）	3DR 915Mhz 1000mw	300	

约23000（不含夹爪，碳板等零件）

电池尺寸：

电芯	EVEINR21700-40PL	EVEINR21700-50PL	EVEINR21700-40PL	EVEINR21700-50PL
容量	4000mAh	5000mAh	8000mAh	10000mAh
电压	22.2V	22.2V	22.2V	22.2V
电芯组合	6S1P	6S1P	6S2P	6S2P
瓦时(Wh)	88.8Wh	111Wh	177.6Wh	222Wh
尺寸(mm)	43*63*74	43*63*75	43*64*147	43*63*148
重量(g)	430±5g	436±5g	840±10g	855±10g
放电倍率	17.5C	25C	17.5C	25C
放电插头	XT60H-F	XT60H-F	XT60H-F	XT60H-F

UAV 配件

水触发救援装置

<https://drone-retriever.com/shop/ols/products/drone-retriever-usa-free-usa-ground-shipping/v/DRN-RTRVR-DR9LFTS9LBS4KG-TSDSNTRNTNLNRDRTPRVDTHLWSTCSTSHPPNGWNLYPRCSSRDRSMNLLYSNDTSLSDRNRTVRRCMXCTLYWHTYWNTTPRCHSYRCMPLTSHPPNGDD>

无人机状态指示灯

https://www.amazon.com/dp/B00NPCR6XE?ref=cm_sw_r_cp_ud_dp_7HZ8QE502VK6T3JMW18V&ref_=cm_sw_r_cp_ud_dp_7HZ8QE502VK6T3JMW18V&social_share=cm_sw_r_cp_ud_dp_7HZ8QE502VK6T3JMW18V

打捞灯（组委会提供）

<https://s.taobao.com/search?bx-ua=234%21txOeKX95eePWvt7pl%2BPeluMFz1yApT8L1d%2FkANYjBul96nvC6Ok6JFgpBu...>
s.taobao.com

- Each UAV must carry an **emergency recovery light** that operates independently from all other platform power systems and is capable of functioning underwater for at least 12 hours. Activation of this light must be demonstrated prior to each launch of the system.
 - At the start of the competition each team with a UAV will be provided with a water-activated flashing light to attach to their UAV.
 - The selected water activated light is available at Amazon: <https://a.co/d/0g1AjKYO> ↗



抓取装置

把夹爪的选型更新在这里

夹爪



三指防水板舵机柔性夹爪

<https://detail.tmall.com/item...>
detail.tmall.com

舵机参数

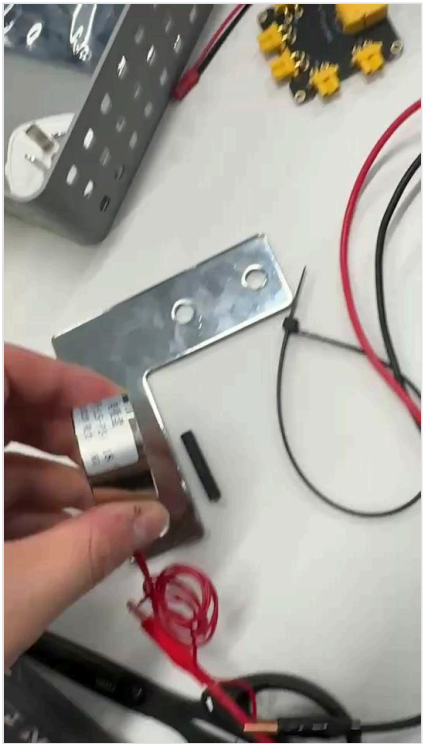
参数	型号	二指舵机	三指舵机	四指舵机	三指舵机 (防水版)	四指舵机 (防水版)
尺寸		76*56*110.8mm	114*45*154mm	115.7*60*154mm	114*45*153mm	115.7*60*153mm
重量		148g	240g	300g	240g	300g
抓取尺寸		15-82mm	15-90mm	5-90mm	20-100mm	5-90mm
抓取重量		0.5kg	0.9kg	1.2kg	0.9kg	1.1kg
供电电压		5V				
抓取频率		<60次/分				

三指舵机（防水版）

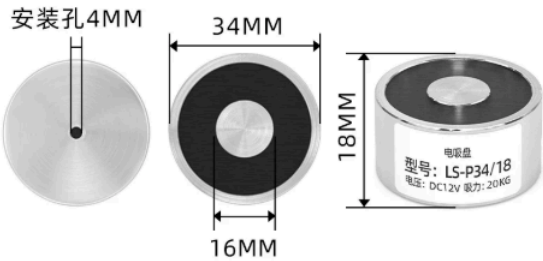
or 电磁铁

【DC12V】LS-P34/18 吸力20KG

<https://detail.tmall.com/item...>
detail.tmall.com



LS-P34/18
电压12V吸力20KG



电压	电流	功率	孔径	吸力
DC12V	0.28A	3W	M4	20KG

or真空吸盘+气泵

370气泵大气量版(12V)

有红点一端是正极，两端焊接正负极后，连接mosfet，另一端连接12V电源线

<https://item.taobao.com/item...>
[item.taobao.com](https://item.taobao.com/item...)

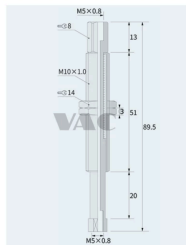
12V气泵

<https://item.taobao.com/item...>
[item.taobao.com](https://item.taobao.com/item...)

吸盘

缓冲杆

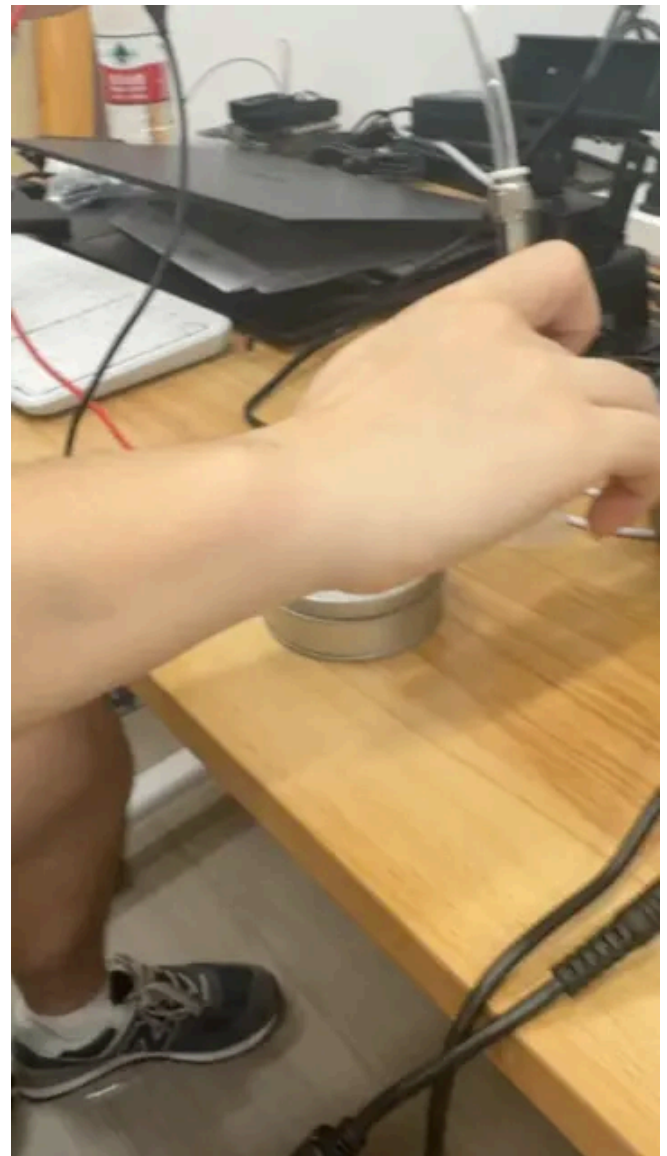
PSPE-I20M5F-M10 (内牙)



VAC® 维爱斯

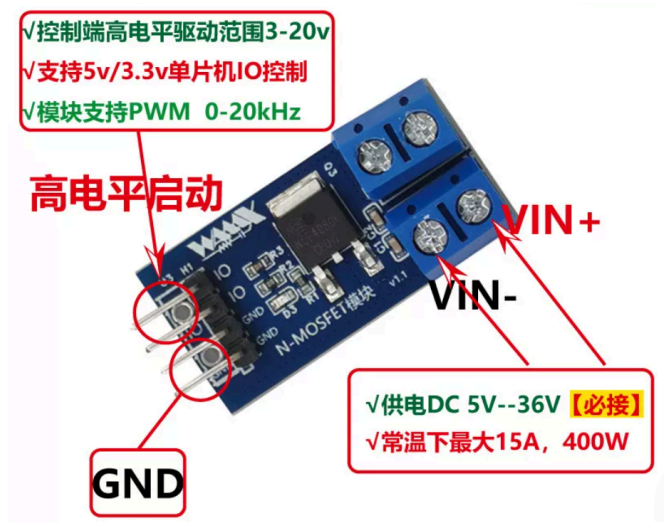
铜镀镍材质

<https://item.taobao.com/item...>
[item.taobao.com](https://item.taobao.com/item...)



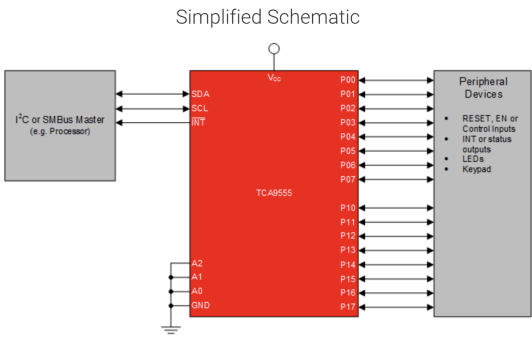
MOSFET

<https://item.taobao.com/item...>
[item.taobao.com](https://item.taobao.com/item...)



TCA9555模块
(I2C转gpio)

<https://detail.tmall.com/item...>
detail.tmall.com



3D打印件

Once finished, pls update

宏烨 马宏烨

STL:

before:

-  云台与摄像机.STL
95.98KB 
-  前下视摄像头固定配件.STL
382.99KB 
-  夹爪与下视固定板.STL
250.47KB 
-  支撑架.STL
61.61KB 
-  支撑架(镜像).STL
61.80KB 

8.12:





前下视摄像头固定配件
v2.0.STL
306.53KB





夹爪与下视固定板v2.0.STL
250.67KB



☒

8.13



前下视摄像头固定配件
v3.0.STL
294.03KB



☒



防水壳.STL
747.93KB



☒

8.14



防水壳v2.0.STL
775.96KB



☒



前下视摄像头固定配件
v4.0.STL
307.31KB



☒

8.15



云台与摄像机v2.0.STL
87.97KB



☒

8.16



TOP_PLATE.STL
693.64KB



☒



前下视摄像头固定配件
v5.0.STL
307.31KB



☒

 赵佳研

8.18



气泵固定件.STL
99.89KB



☒

- 

电磁铁固定件.STL
 77.23KB


☒
- 

云台底板.stl
 87.19KB


☒

9.9

- 

前下视摄像头固定配件
 v6.0.STL
 316.29KB


☒
- 

云台与摄像机v3.0.STL
 102.43KB


☒

9.12

- 

前下视摄像头固定配件
 v6.2.STL
 316.29KB


☒

9.16

- 

云台与摄像机v3.1.STL
 102.23KB

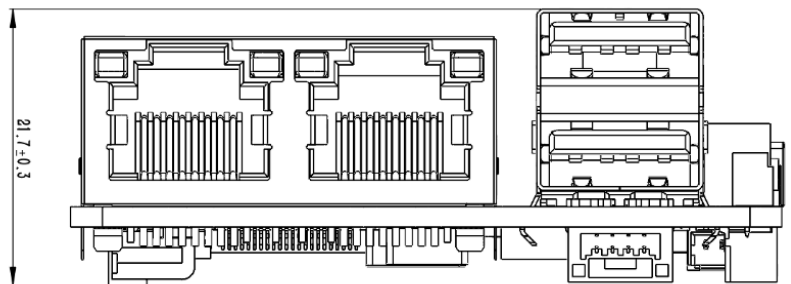
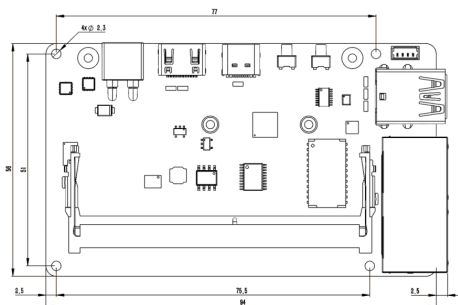

☐

尺寸、引脚定义：

Pixhawk 6x mini baseboard pin definition:

<https://docs.holybro.com/autopilot/pixhawk-baseboards/pixhawk-mini-baseboard-ports>

On board computer 尺寸：



加工工具清单

ppr水管剪刀pvc割刀快剪波纹管子割刀线管剪管钳

Light Beacon 物料清单

<https://robonation.gitbook.io/robotx-2026-team-handbook/robotx-2026-or-task-build-guides?fallback=true>



单个Beacon：

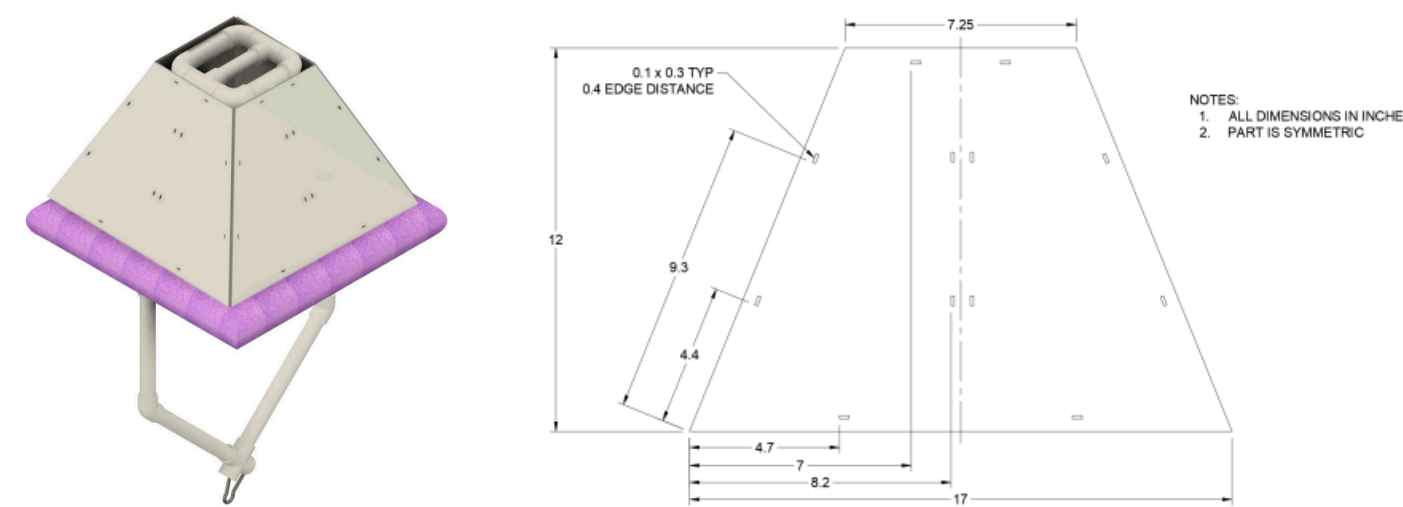
物品	数量	链接
2m IP67 LED strip	1	
8*8 WS2812B led matrix	1	
Microcontroller ESP32 Dev	1	
12V LiFePO4 （12V 21700 Li-ion）	1	
杜邦线	若干	
Battery checker	1	
防水盒子125mm x 125mm x 75mm	1	

防水插头 (LED cable - inner)	1 - 2	
DC-DC convertor (12V - 5V)	1	

测试需要6组

Robo buoy 物料清单

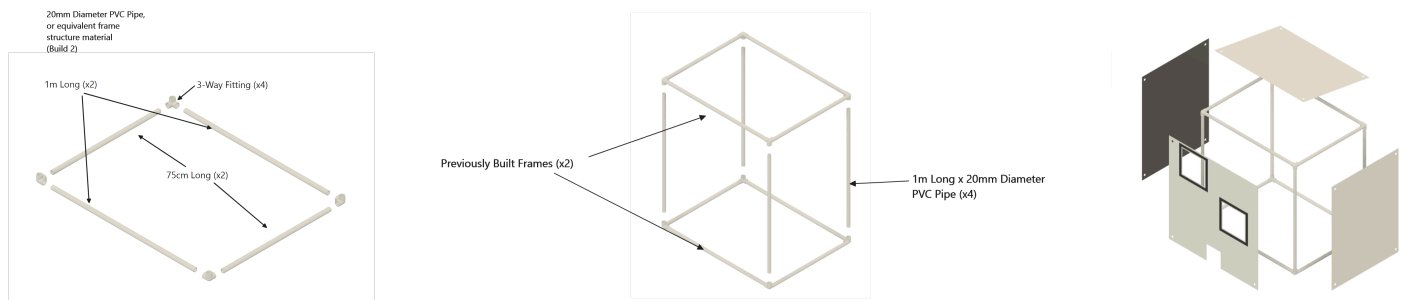
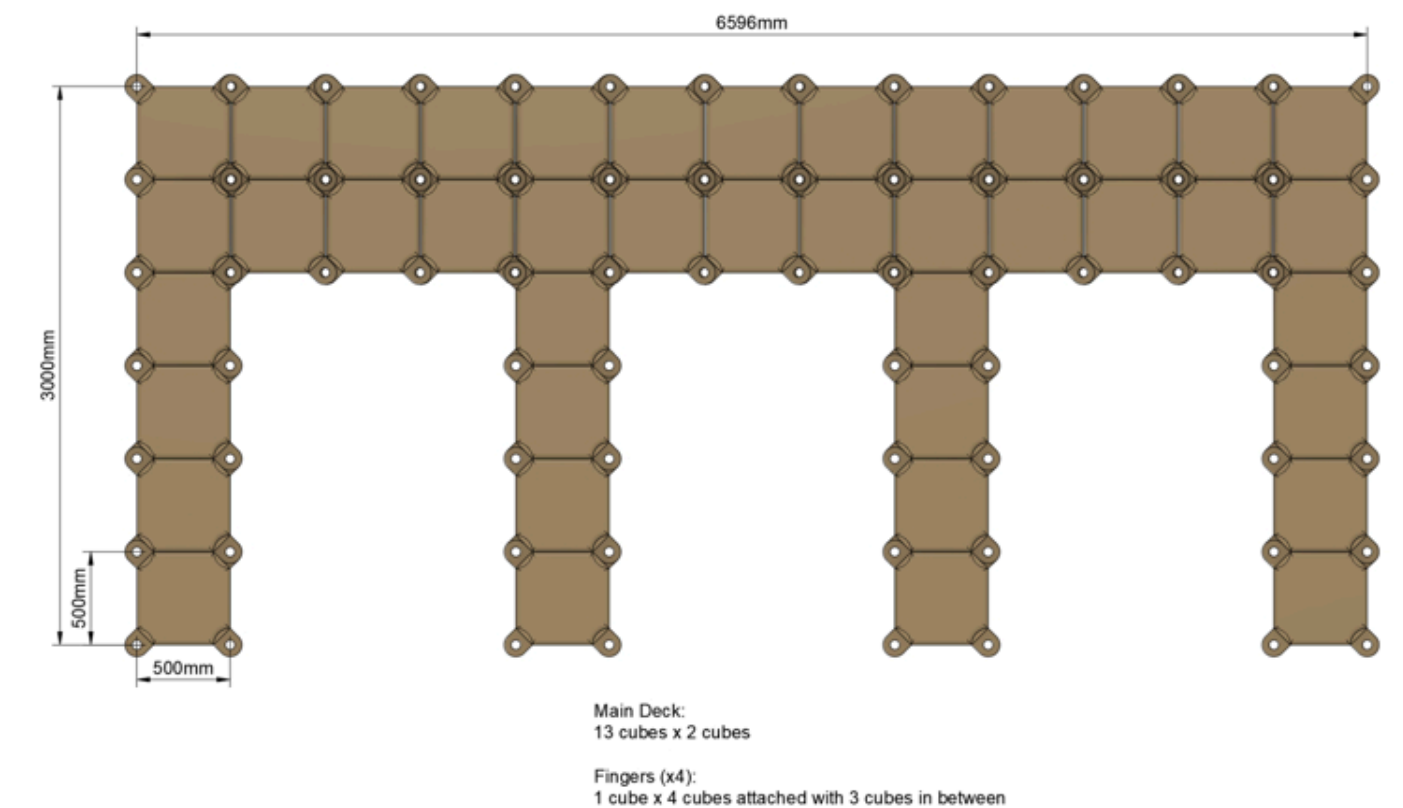
<https://robonation.gitbook.io/robotx-2026-team-handbook/robotx-2026-or-task-build-guides?fallback=true>



物品	数量	链接
PVC sch40 (1m)	5	
白色瓦楞塑料板 (1m*1m) PP塑料板	1	
浮力材料(1m)	2	
三通sch40	6	
四通	1	
PVC sch40 45度	2	
PVC sch40 直角	8	
Spears 463 系列 3/4 in Snap-On Saddle (3D打印)	6	

Docking Bay Structure 物料清单

<https://robonation.gitbook.io/robotx-2026-team-handbook/robotx-2026-or-task-build-guides?fallback=true>



物品	数量	链接
PVC 20mm （1m）	8	
PVC 20mm （0.8m）	4	
PVC 20mm 三通	8	
白色瓦楞塑料板（1m*1m）PP塑料板	4	

Color Indicator 物料清单

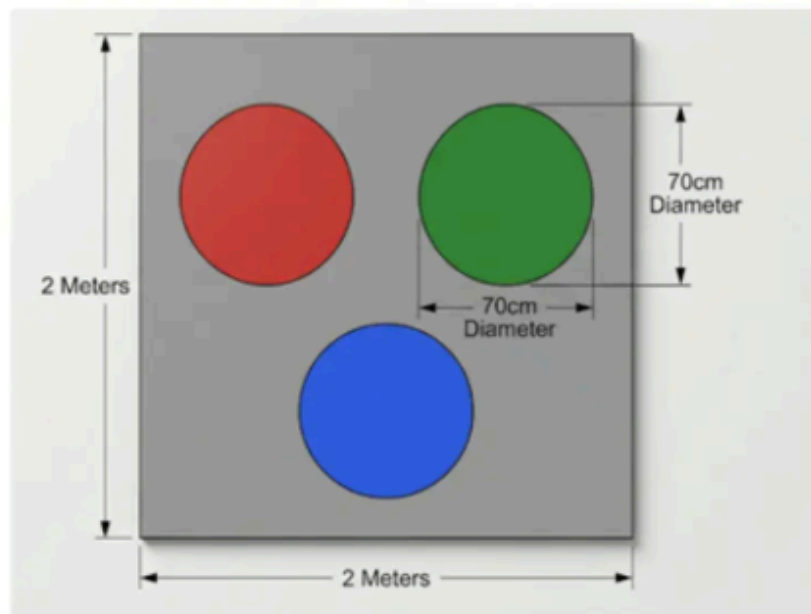
<https://robonation.gitbook.io/robotx-2026-team-handbook/robotx-2026-or-task-build-guides?fallback=true>



物品	数量	链接
盖子 (3D打印)	1	
红色fins (3D打印)	1	
底板+绿色fins (3D打印)	1	

Landing Pad & Delievery Platform 物料清单

There will also be two floating platforms within the course boundary that contain **RED**, **GREEN**, and **BLUE** tins. Each of the floating platforms will be approximately 2 meters square, covered with a grey background with **RED**, **GREEN**, and **BLUE** circles on it. The circles will be 60-70cm in diameter. These platforms are also used in Task 3.



Example UAV Delivery Platform

The colors for the background and colored circles will be as near as we are able to produce to those listed in the table below.

Platform Element	Hex Code	Pantone Color
Background Color (Benjamin Moore Sidewalk Grey ↗)	#CCCED0	179-3 C
Red Circle	#E8282B	1788 C
Green Circle	#089404	355 C
Blue Circle	#0072CE	285 C

Tins 物料清单

https://item.taobao.com/item.htm?id=846309125052&mi_id=0000PhtbdakpOv8zMJJ6Lii-VZ46YwvUbdG_95nVyFnL02M&sku_properties=21433%3A605190186&spm=tbpc.boughtlist.suborder_itemtitle.1.32f62e8dts5r6w



通用银色圆形开窗马口铁盒 茶叶盒蛋糕烘焙礼品铁盒子 88元包邮 [交易快照]
直径6.5*高2.5CM - 开窗 - 银色;平盖款
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Tins 颜色

户外实验物料清单

物品	数量	链接
桌子120cm*60cm	1	
椅子	4	
遮阳棚17m^2	1	
户外推车	1	
拖拽杆	2	
移动电源	1	